

First Name

Functional Programming Written Test

Quiz Bank Parser and Paper Generator

Duration: 1.5 hours

Total Marks: 65 marks

Question	Topic	Marks
1	Modelling the Domain with ADTs	10
2	Parsing XML into Haskell Values	9
3	Rendering Markdown Output	6
4	Invariants and Validation	5
5	Valid Paper Generation	8
6	QuickCheck Properties	8
7	Category Inheritance	5
8	Program Structure and Design	9
9	Reflection on using AI Bots	5
	Total	65

Instructions

- Answer all questions.
- You do not need to write a complete working program. Where code is requested, clear Haskell-style code or well-structured pseudocode is acceptable unless otherwise stated.

Name	Student Number

Question 1 - Modelling the Domain with ADTs (10 marks)

In the assignment, each Moodle-style XML quiz file is parsed into an internal Haskell representation.

- a)** Define suitable Haskell data types to represent: a quiz/question bank, a question, many possible answers, a category, and marks per question. (Slightly different than the one in your assignment)

[6 marks]

ADT's

- b)** Briefly explain two advantages of using algebraic data types rather than storing everything as raw String values.

[4 marks]

advantages	
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Question 2 - Parsing XML into Haskell Values (9 marks)

Consider the following simplified XML fragment:

```
<quiz>
  <category>Loops</category>
  <question type="multichoice">
    <name>While loop question</name>
    <questiontext>
      What does a while loop do?
    </questiontext>

    <answer fraction="100">
      <text>Repeats while a condition is true</text>
      <feedback>Correct. A while loop continues while its condition evaluates to true.
      </feedback>
    </answer>

    <answer fraction="0">
      <text>Runs exactly once</text>
      <feedback>Incorrect. A while loop may run zero, one, or many times.</feedback>
    </answer>

    <answer fraction="0">
      <text>Repeats a fixed number of times only</text>
      <feedback>Incorrect. A for loop is usually more suitable for a fixed num of repetitions.
</feedback>
    </answer>

    <answer fraction="0">
      <text>Stores a list of values</text>
      <feedback>Incorrect. A loop controls repetition; it does not store values.</feedback>
    </answer>

    <answer fraction="0">
      <text>Defines a new data type</text>
      <feedback>Incorrect. Defining data types is separate from using loops.</feedback>
    </answer>
  </question>
</quiz>
</quiz>
```

a) Draw or describe the tree structure of this XML fragment

[3 marks]

Tree structure

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b) Explain how a parser should extract the quiz category, the question text, and the answer text.

[6 marks]

Extracting data

Question 3 - Rendering Markdown Output (6 marks)

Using the ADT's that you wrote in Question 1, and assuming you have a list of 'questions', show how you would write Markdown output for:

a) Just questions (no answers).

[3 marks]

Just questions

[3 marks]

Questions and answers

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Question 4 - Invariants and Validation (5 marks)

The assignment requires students to identify and enforce invariants.

- a) Give three important invariants for the question bank (using the(your)assignment details).

[3 marks]

Question Bank -invariants

- b) Give 2 invariants for the generation of a generated 'exam paper'(using the(your)assignment details).

[2 marks]

Exam paper invariants

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Question 5 - Valid Paper Generation (8 marks)

A lecturer wants to generate a paper with the following constraints:

At least 2 questions from category "Loops"
At least 1 question from category "Lists"
Total number of questions must be 10

a) Explain what your program needs to check before returning a generated paper.

[3 marks]

Program needs

b) If the question bank cannot satisfy these constraints, what should the program do?

[2 marks]

What should program do

c) Suppose a function has this type:

```
generatePaper:: PaperSpec -> QuestionBank -> Either String Paper
```

Explain why Either String Paper is a better return type than just Paper.

[3 marks]

Why Either

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3

Question 6 - QuickCheck Properties (8 marks)

QuickCheck is used to test important properties of the solution.

- a) Write or describe a QuickCheck property for this rule:

Shuffling preserves the number of questions.

[2 marks]

Shuffling

- b) Write or describe a QuickCheck property for this rule:

If a paper is generated successfully, it has the required total number of questions.

[3 marks]

Total

- c) Describe how you would use stack to run the properties (show property and the QuickCheck call). How do you run the property checks from the terminal?

[3 marks]

running

Question 7 - Category Inheritance (5 marks)

Each XML file has a default quiz category. Some questions may also have their own explicit category.

The assignment rule is: if a question has an explicit category, use it; otherwise, inherit the default quiz category.

Show how you developed your solution for this. Write a Haskell function, or Haskell-style pseudocode, that deals with this ‘specific or default’ issue.

[5 marks

Total	
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Question 8 - Program Structure and Design (9 marks)

a) A good solution separates responsibilities into different modules, for example:

```
Adts.hs
Parse.hs
Validate.hs
RenderMarkdown.hs
GeneratePaper.hs
Main.hs
```

Explain what each of these modules might contain, and why this structure is better than putting the whole program into Main.hs.

[5 marks]

Module contents

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- b) Explain how your (own, assignment) design helped you to generate a (test) paper with minimum numbers of given categories.

[4 marks]

Category design

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Question 9 - Reflection on using AI Bots (5 marks)

If you used an AI Bot to write your code, give an example of

1. How you generated the help (what kind of prompts)
2. How you needed to modify the generated code (and why)
3. How you improved the code by this intervention.

AI